

Unlocking Autistic Children's Potential: The Crux with Data Visualisations and IoT

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ABSTRACT

Internet of Things (IoT) technology and data visualisations have significant potential to support autistic children in their self-awareness, self-regulation, learning, and daily activities. IoT applications collect extensive data on behaviours, activities, and environmental and contextual factors. Traditionally, this data is often visualised to create more or less meaningful and actionable insights for the users. This paper examines the current landscape of data visualisations used for autistic children, identifying a significant gap in understanding how autistic children interact with these data visualisations. By highlighting the limitations of existing tools and guidelines, this work lays the groundwork for future experimental research aimed at developing tailored visualisation strategies and enhancing visualisation literacy in autistic children. Insights from this work advance the knowledge of autistic children's specific needs, ultimately contributing to designing more effective and inclusive visual interfaces to enhance their self-awareness, learning, and interventions.

CCS CONCEPTS

• **Human-centered computing** → **Visualisation; Interaction design; Accessibility.**

KEYWORDS

Autism; Data Visualisation; IoT technology; Accessibility

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1 INTRODUCTION AND BACKGROUND

Autism Spectrum Disorder (ASD), as defined by the American Psychiatric Association in the Diagnostic and Statistical Manual of Mental Disorders fifth edition (DSM-5), is a neurodevelopmental disorder characterised by deficits in social communication and interaction across multiple contexts, as well as restricted, repetitive patterns of behaviour, interests, or activities [5]. The prevalence of ASD is increasing globally, affecting roughly 1 in 100 children [37].

Besides social and communication difficulties, autistic individuals often exhibit altered sensory reactivity, which can manifest as hypersensitivity (extreme sensitivity to stimuli like light or sound) or hyposensitivity (reduced response to stimuli, such as higher pain and temperature thresholds). These sensory particularities show significant variability among individuals, affecting about 90% of autistic people [13, 26]. Visual perceptual differences are particularly significant, with sensory abnormalities observed in approximately 95% of autistic children [9]. These differences can lead to challenges in interpreting visual information, posing barriers to effective communication and learning.

Moreover, many autistic individuals face challenges in monitoring and being aware of their own emotional states. This is attributed to a reduced ability to sense and interpret internal physiological cues [19]. Emotional self-awareness difficulties are more common in autistic individuals and typically exacerbate with age, analogous to the progression of mental health issues within this population, often starting during adolescence [17].

Besides the general increase in Internet of Things (IoT) device usage, IoT technologies are also increasingly used to support autistic children. IoT applications employ sensors and devices to monitor and collect data on behaviours, activities, and environmental factors [4, 11]. These applications, functioning as *intimate user interfaces* hold substantial potential for advancing into an empowerment paradigm. This paradigm represents an opportunity to view autistic children as autonomous and self-determined users, rather than as subjects requiring intervention based on medical pathology [32]. By providing accessible visualisations that they can interact with directly, IoT technologies support independence, enhance self-regulation, and reduce reliance on caregivers.

Data visualisations, including statistical charts, diagrams, and maps, are widely used to foster understanding, communicate complex information and identify insights. They leverage the human visual system's capacity for high bandwidth, parallel processing, and rapid pattern recognition [18]. For example, a line chart can reveal trends more clearly than a table of numbers. This underscores the potential of personal informatics in developing self-regulation skills [23]. There are clear opportunities for these tools to be made accessible to autistic children to help them manage their everyday lives, e.g., their health and emotions autonomously [3]. Furthermore, these visualisations also enable caregivers and therapists to use the data for making informed decisions about future therapies and provide evidence of intervention effectiveness [16].

Understanding and interpreting data visualisations requires substantial cognitive effort, including quantitative reasoning, statistical estimation, and information retention [2, 25]. These processes often function differently for individuals with cognitive and learning disabilities [34]. Moreover, the design of these visualisations frequently assumes typical sensory, cognitive, and motor abilities [18]. Consequently, those lacking these abilities may be excluded, limiting access to the benefits of data visualisations and the underlying information. This exclusion is particularly challenging for autistic children, who often face unique difficulties in interpreting abstract concepts due to their visual processing issues [9]. It is important to recognise that disability, as a natural part of the human experience, does not exclude individuals from living independently, enjoying self-determination, and making decisions, as explicitly declared by the Rehabilitation Act of 1973 [22]. Thus, ensuring accessible data visualisations is crucial for empowering autistic children and supporting their autonomy.

Integrating IoT applications and data visualisations promises to enhance understanding, self-regulation, and independence and reduce reliance on caregivers, ultimately improving the quality of life for autistic children. Despite the recognised importance of IoT and data visualisations, there is a lack of research on whether autistic children can understand and effectively use these visualisations. Our work aims to improve our understanding of how visualisations can empower autistic children to engage with their data meaningfully and how they can be integrated into educational settings to support learning and self-regulation.

2 IOT AND DATA VISUALISATION FOR AUTISTIC CHILDREN

In this section, we examine visual processing and self-awareness in autism, highlighting the unique challenges faced by autistic individuals. We then review studies demonstrating the importance of understanding health and personal data from IoT technology for enhancing self-awareness and improving overall health management. Notably, most IoT technology designs are tailored for adults, often overlooking the specific needs of children. We present studies that explored using IoT technology to enhance interventions for children through data sharing with their families, these studies typically focus on neurotypical children. Finally, we review studies where IoT technology has been employed to support autistic children, noting that visualisations are primarily designed for caregivers and parents rather than for the children themselves.

2.1 Visual Processing and Self-awareness in Autism

Visual processing difficulties are commonly observed in individuals diagnosed with autism, impacting their ability to interpret and respond to visual stimuli. Research has shown that autistic individuals often experience heightened sensitivity to certain visual features, such as brightness, colour, and contrast, which can lead to sensory overload and discomfort [30]. Additionally, they often struggle with visual integration, affecting their ability to recognise faces, read emotions, and navigate environments, typically focusing on details rather than the whole picture, a phenomenon known as *weak central coherence* [15].

In addition to visual processing challenges, many autistic individuals face difficulties in monitoring and being aware of their own emotional states. This impairment is due to their reduced ability to sense and interpret internal physiological cues [19]. Emotional self-awareness difficulties are more prevalent in autistic individuals and tend to worsen with age, similar to the development of mental health issues. Notably, these differences in emotional self-awareness typically begin to emerge during adolescence [17]. Understanding these sensory and emotional self-awareness issues is crucial for designing effective data visualisations that cater to the needs of autistic children. By leveraging IoT technology to create accessible visualisations, we can enhance their ability to understand and manage their data, improving their emotional self-awareness, independence, and overall quality of life.

2.2 IoT Technologies and Data Visualisation

IoT technology encompasses a network of interconnected devices that collect and exchange data through the internet, facilitating the real-time monitoring and management of various aspects of daily life [11]. To make this data useful, it is crucial to process and present it to the user, e.g., by visualising it in an understandable and actionable manner. Data visualisation effectively simplifies complex data sets, making interpreting and acting upon the information easier [6]. By transforming raw data into visual formats such as charts, graphs, and maps, data visualisations can reveal patterns, trends, and correlations that might be missed in text-based data sets. This enables individuals to quickly grasp key insights and underlying trends, which is particularly important for those without a technical background [7].

2.2.1 Enhancing Self-Awareness and Self-Monitoring Through IoT Data Visualisation. Visualising data is a powerful tool for transforming IoT data into meaningful insights that can help individuals comprehend personal and health information, aid in self-reflection and self-awareness, and support self-monitoring [10, 23]. Whether this information is shared or kept private, individuals who engage in self-monitoring may experience changes in their behaviour simply by paying attention to it [24]. In an era increasingly driven by data, deciphering and engaging with data visualisations is crucial, providing everyone, including autistic children, with the tools to better understand and manage their lives.

However, there are notable limitations in existing research and designs. Primarily, the focus tends to be on creating technologies

and visualisations for the general population, often neglecting the specific needs of children, particularly those with autism.

2.2.2 Enhancing Interventions Through IoT Data Visualisation. Traditionally, visualisations and interventions for children have relied heavily on collaboration with their caregivers. As early as 1999, researchers in family medicine and other fields recognised the need for a different approach to data visualisations in health technologies [12]. Recently, there has been a push to adopt an ecological perspective on child interventions, formalised as family informatics [16]. For example, Pina et al. [21] introduced *DreamCatcher*, a tool that allows families to collaboratively track sleep and mood using wrist-worn sleep trackers and an in-home display. The authors argued that existing sleep-tracking tools were adult-centric and not suitable for family use, demonstrating how children can be active participants in tracking and collaborating with their parents. Similarly, Saksono et al. [27] evaluated *Storywell*, a system that integrates wristband activity tracking for parents and children with social rewards in the form of storybooks. These storybooks include graphical and narrative elements that encourage goal setting and reflection, making the tracking process more engaging and meaningful for children. Müller et al. [20] assessed the feasibility of wearable bracelets for influencing physical activity levels in children, finding them suitable for assessing physical activity but raising concerns about young children's ability to comprehend the displayed activity metrics. Ananthanarayan et al. [1] noted that children track and interpret health data differently than adults, suggesting that health data visualisations should be specifically designed for children. Further research is required to determine the extent to which these differences affect the experience of viewing visualisations, especially for autistic children. However, it is important to note that these studies primarily involve neurotypical children, and their findings may not fully address the needs of autistic children, and hence there could be a need for further research focused specifically on this population.

2.2.3 Enhancing Interventions Through IoT Data Visualisation for Autistic Children. IoT technologies play a crucial role in recognising and analysing the emotional states and effectiveness of therapeutic interventions for autistic children. For example, the Autism app/framework *AutBot* integrates IoT and Artificial Intelligence (AI) technologies to assist in treating autistic children by recognising their emotions and analysing their performance during prescribed activities. It helps doctors and therapists use data from this tool to efficiently analyse the emotional states and behaviours of autistic children, thereby enhancing the effectiveness of their treatment plans [29]. Moreover, IoT systems that monitor mental states through deep learning algorithms for emotion recognition have shown promise in supporting autistic children by improving the understanding of their emotional states and aiding in communication. Recent advancements include using models such as ER16Net and AlexNet to analyse body movements and recognise emotions in autistic children. These deep learning algorithms have proven effective in extracting deep features and providing valuable insights into the emotional states of these children [28]. Additionally, Hayes et al. [16] explore the design of automated capture technologies that can record and visualise valuable data from various interventions. These visualisations allow caregivers to quickly understand overall

trends and detailed views, informing decisions about structuring future therapies. Such visualisation tools are indispensable for ensuring the data collected through automated capture technologies is accessible and actionable.

While these advancements offer valuable insights into the emotional states and behaviours of autistic children, as well as into interventions, they are primarily designed using interfaces and visualisations for caregivers and doctors, thereby neglecting the specific needs of autistic children. For autistic children, understanding and interacting with their own data is crucial, as it can significantly reduce dependence on caregivers, empower them to manage their health and emotions more independently, and promote greater autonomy and self-regulation [3].

3 AUTISM AND DATA VISUALISATION

Data visualisations have the potential to significantly empower the autistic cohort by making personal information more accessible and understandable. Research indicates that individuals with Intellectual and Developmental Disabilities (IDD), including autism, frequently utilise data for various purposes, such as advocating for their needs, sharing personal experiences, and expressing creativity [36]. Accessible visualisations assist in monitoring personal health, behaviours, and progress, thereby promoting autonomy [35]. However, many current data visualisation tools do not adequately address the unique sensory and cognitive needs of autistic individuals, thereby limiting their access to essential information [33, 34].

A compelling example of the value of tailored data visualisations is the Personalised Interactive Urban Maps for Autism (PIUMA) project [8]. This initiative aims to assist autistic individuals in navigating urban environments by offering personalised maps that cater to their specific cognitive and sensory needs. The PIUMA project underscores the importance of visual tools in supporting the autonomy of autistic individuals, enabling them to plan and manage everyday movements, locate suitable points of interest, and handle unexpected events with greater ease.

Moreover, data visualisations can enhance the learning experiences of children with cognitive disabilities. For instance, research in education by Zhang et al. [39] demonstrated that the use of visual-chunking representation significantly improved geometry problem-solving performance in students with math disabilities. Visual-chunking groups individual visual elements into larger chunks, reducing the cognitive load on visual working memory. The study used an adapted reversal design (alternating baseline and intervention phases) to compare performance under standard conditions and with visual-chunking. Four third-grade students with math difficulties showed significant improvement, with accuracy increasing from 46% to 78%. Interviews confirmed a preference for chunked visual representations. Additionally, Van Garderen's [31] study highlighted the effectiveness of using schematic imagery, which represents spatial relationships and abstract concepts, in improving mathematical problem-solving among students with varying abilities. These findings highlight the potential for data visualisations to render abstract information more tangible and comprehensible.

However, current research on how children with cognitive disabilities understand and interact with their data is limited. Ankrah et al. [3] conducted a study with 10 children with Attention Deficit Hyperactivity Disorder (ADHD), examining how they interpret and make meaning of their health data from smartwatches over six weeks. ADHD children often face challenges related to self-regulation, leading to poor educational, health, and wellness outcomes. The study involved the children interacting with health data like physical activity levels, emotions, and other default applications on the smartwatches. The visualisations presented to the children include the available smartwatch interfaces and the *CoolCraig* application. The *CoolCraig* application was developed by the research team, teachers, and other school staff to help children self-report their emotions using visual cues based on the Zones of Regulation framework. This app prompted children to answer "How are you feeling?" three times a day using emojis with different background colours (green, blue, red, yellow) and secondary visual cues to indicate their emotions. While the children could understand their data in real time and found that the devices supported their autonomy, they often struggled to derive meaningful insights from the available smartwatch visualisations. They frequently used the data to reinforce pre-existing beliefs, particularly concerning the relationship between their health data and emotions. Despite these challenges, the use of smartwatches allowed the children to exercise autonomy in managing their health and emotions, underscoring the potential of personal informatics to foster self-regulation skills. The study highlights significant challenges and gaps in current data visualisation guidelines for ADHD children, emphasising the need for tailored visualisation strategies that can better support meaningful interpretation and emotional regulation.

Wu et al. [34] found that individuals with IDD and autism employ different strategies to interpret data, indicating this group has traditionally been ignored in visualisation design and often has limited access to data related to their well-being. Thus, common visualisation guidelines can create barriers for them. The study identified three key visualisation design elements—chart type, chart embellishment, and data continuity—that can improve data accessibility for this population. For instance, avoiding pie charts, using natural metaphors, and supporting working memory with simple imagery can enhance comprehension and usability. These findings suggest that the cognitive processes and visualisation needs of autistic children are not adequately addressed by existing tools. In addition, Wimer et al. [33] conducted a systematic review of 152 accessible data visualisation papers from ACM and IEEE databases, discovering that approximately 78% of these articles focus on vision impairments. Their comprehensive analysis of the remaining 22% of papers, of which only 3% addressed cognitive disabilities, revealed significant gaps and opportunities for future research into accessible data visualisation. To the best of our knowledge, there is a lack of data visualisation research specifically aimed at autistic or neurodiverse communities beyond the studies mentioned above. This highlights a significant need for more focused research in this area to develop effective and inclusive visualisation tools that cater to the unique needs of these populations.

4 OPPORTUNITIES, BENEFITS, AND CHALLENGES

Integrating IoT applications and data visualisations offers numerous opportunities and benefits for autistic children, including enhanced self-awareness, autonomy, and quality of life. Moreover, these technologies can create highly personalised interventions tailored to each child's needs, enhancing their care. In educational settings, integrating these tools can make learning more interactive and effective, providing new ways to engage autistic children. For example, Zhai et al. [38] present an interactive educational framework using a Machine Learning-based Virtual Reality Application (ML-VRA) for autistic children, which enables them to effectively acquire various levels of knowledge and skills. Hailpern et al. [14] noted that interactive visualisations could help teach speech and language skills to autistic children through visual and auditory feedback. Additionally, these tools offer valuable insights to caregivers and therapists, facilitating improved data-driven decision-making for more effective treatment plans. Furthermore, IoT technology and data visualisations empower autistic children to monitor and manage their own health and behaviours, fostering greater independence.

Incorporating user-centred design principles in data visualisation studies for autistic individuals presents significant opportunities to create accessible and effective solutions. Engaging autistic children and their caregivers throughout the research process is crucial for understanding their needs and preferences [18], leading to the development of tools that support their autonomy and well-being. Additionally, involving neurodiverse individuals in the research process enhances the accessibility and effectiveness of the visualisations, ensuring the tools are designed with a comprehensive understanding of the community's unique challenges and strengths.

However, conducting research with autistic children presents several challenges. Recruitment can be difficult due to factors such as obtaining parental consent, availability, and willingness to participate. The individual variability in abilities and sensory sensitivities among autistic children further complicates this process. Finally, communication barriers may also hinder accurately gathering feedback on their experiences with data visualisations.

5 CONCLUSION

IoT data visualisation holds immense potential for enhancing the lives of autistic children by improving their self-awareness, self-regulation, learning, and daily activities. These visualisations can transform extensive data on behaviours, activities, and environmental factors into meaningful and actionable insights tailored to the unique sensory and cognitive needs of autistic children. Despite this potential, there is a clear need for more research to understand how autistic children interact with and interpret these visualisations and to develop effective visualisation strategies that specifically address the challenges they face. Tailored visualisations have the potential to significantly enhance the autonomy and quality of life for autistic children by making health and behavioural data more accessible and understandable. This can reduce their reliance on caregivers, foster greater independence, and support more effective learning and therapeutic interventions. We urge researchers and designers to prioritise creating inclusive and accessible data visualisation tools

that cater to the needs of autistic cohorts. By focusing on these specific requirements, the field can make substantial strides towards more empathetic and effective technological solutions, ultimately promoting greater inclusion and empowerment.

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